Qn. Set Code-7(SOT)

Semester: 5th
Programme: B.Tech
Branch: Comp. Engg.

AUTUMN END SEMESTER EXAMINATION-2025

5th Semester B.Tech

MACHINE LEARNING

CS3035

(For 2022 & Previous Admitted Batches)

Time: 2 Hours 30 Minutes | Full Marks: 50

Answer any FIVE questions.
Question paper consists of two SECTIONS i.e. A and B.
Section A is compulsory.
Attempt any Four question from Sections B.
The figures in the margin indicate full marks.
Candidates are required to give their answers in their own words as far as practicable and all parts of a question should be answered at one place only.

SECTION-A

1. Answer the following questions. [1 × 10]

(a) Choose the correct answer:

The empirical cost function in supervised learning is calculated taking into

A. Training samples only
B. Testing samples only
C. Both A and B
D. Either of A or B depending on the application/data distribution.

(b) Standardize the given dataset of 1D points: $D = \{5, 10, 15\}$.

(c) Choose the correct answer:

In a Support Vector Machine (SVM), suppose the data is linearly separable, and all input features are scaled by a constant factor C (> 1). How does this scaling affect the optimal weight vector w ∗ and the resulting margin?

A. w ∗ remains the same, and the margin increases by a factor of C

B. w ∗ decreases by a factor of C, and the margin increases by a factor of C

C. w ∗ increases by a factor of C, and the margin decreases by a factor of C

D. Both w ∗ and the margin remain unchanged.

(d) State whether hierarchical clustering requires specifying the number of clusters in advance and provide a justification.

(e) Define the curse of dimensionality and explain its negative impact on the performance of distance-based machine learning algorithms like K-Nearest Neighbors (KNN).

(f) Describe the role of the Learning rate in the training of a neural network.

(g) Explain the usefulness of KKT conditions.

(h) In a binary classification problem, the confusion matrix has the following values:

True Positives (TP) = 125, and True Negatives (TN) = 35. If the model's accuracy is 80% and precision equals recall, determine the values of False Positives (FP) and False Negatives (FN).

(i) State whether the following statement is TRUE/ FALSE, and justify your answer:
Low bias in a model usually leads to overfitting, while high variance usually leads to underfitting.

(j) Explain the importance of bias in a neural network.

SECTION-B

2. (a) Differentiate between Supervised, Unsupervised, and Semi-supervised learning. Explain with an example how Unsupervised learning helps in discovering hidden patterns in data. [5]

(b) How a Multi-layer perceptron (MLP) solves the XOR problem. Design an MLP withback propagation to implement the XOR Boolean function. [5]

3. (a) Differentiate between linear regression and logistic regression with examples. Explain the role of the log-likelihood function in logistic regression training. [5]

(b) Apply the Principal component analysis (PCA) algorithm to the following observations and compute the covariance matrix and eigenvalues. [5]

Data Points	P1	P2	P3	P4
Feature x1	6	12	7	9
Feature x2	8	10	4	12

4. (a) First, normalize all the features of the following dataset. Then, perform K-NN classification to predict the class of a sample point with Engine size = 3.5 and Horsepower = 300, using K = 3 and the Euclidean distance metric. [5]

Sample no.	Engine size	Horsepower	Class label
1	3.2	260	Sports
2	1.4	100	Sedan
3	3.4	280	Sports
4	2.8	210	Sports
5	5.0	460	Luxury Sports
6	1.6	110	Sedan
7	1.8	130	Sedan
8	2.8	220	Sports
9	3.6	310	Sports
10	4.5	410	Luxury Sports

(b) Consider a simple linear regression model $f(x) = a + bx$, the cost function $J(a,b) = \frac{1}{2m}\sum_{i=1}^{m}(f(x_i) - y_i)^2$, and the following dataset [5]

x	1	2	4
y	2	3	5

Use Gradient descent to update the parameters a and b with the following settings:
Initial values: $a = 1.0, b = 0.5$; and learning rate: $\alpha = 0.1$.

Compute the updated values of a and b after the first iteration.

5. (a) Explain how the DBSCAN algorithm identifies clusters of arbitrary shape and distinguishes between core points, border points, and noise points. Discuss the role of the parameters ε (epsilon) and MinPts, and describe how varying these parameters affects the resulting clusters. [5]

(b) A telecom company wants to predict whether a customer will renew their mobile plan based on past behavior. Apply the ID3 decision tree algorithm and calculate the Renewal type if on the following dataset with query (Data uses = Low, Contract type = Annual, Payment timelines = On-time). [5]

Customer	Data uses	Contract type	Payment timelines	Renewal
C1	High	Annual	On-time	Yes
C2	High	Monthly	Late	No
C3	Medium	Monthly	On-time	Yes
C4	Low	Annual	Late	No
C5	Medium	Annual	On-time	Yes
C6	Low	Monthly	On-time	Yes
C7	High	Annual	Late	Yes
C8	Medium	Monthly	Late	No
C9	Low	Annual	On-time	Yes
C10	High	Monthly	On-time	Yes

6. (a) State the characteristics of an artificial neural network. Construct a feed-forward neural network with four input nodes, three hidden nodes, and two output nodes with necessary mathematical expressions and explanation. [5]

(b) Apply the Single linkage hierarchical clustering method to cluster the 2D dataset S = {(2,10), (2,5), (8,4), (5,8), (7,5)}. Use the Euclidean distance to merging step by step until only one cluster remains. Additionally, draw the corresponding dendrogram showing the order of cluster merging. [5]

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